Concentration Inequalities

The Transportation Method

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June 25, 2024

Outline

definitions and results revisit

The Transportation Method

Talagrand's Gaussian Transportation Inequality

The Entropy Method

Entropy

Definition 1 (entropy).

$\Phi(x) = x \log x$ *on* $[0, +\infty)$. *Let* $(\Omega, \mathcal{A}, \mathcal{P})$ *be a probability space and let* Y *be a nonnegative random variable defined on it s.t.* Y *is integrable. The entropy of* Y *is*

$$Ent(Y) = \mathbb{E}[\Phi(Y)] - \Phi(\mathbb{E}[Y]).$$

Definition 2 (relative entropy or Kullback-Leibler divergence).

Y *is a nonnegative random variable with* $\mathbb{E}[Y] = 1$, *we may define another probability measure* $\mathcal{Q}$ *on* $(\Omega, \mathcal{A})$ *by* $\mathcal{Q}(A) = \int_A Y(\omega) d\mathcal{P}(\omega) = \mathbb{E}[Y \mathbb{1}_{\{A\}}]$ *for all* $A \in \mathcal{A}$. *The relative entropy of* $\mathcal{Q}$ *with respect to* $\mathcal{P}$ *is defined by* $D(\mathcal{Q} \parallel \mathcal{P}) = Ent(Y)$.

Note that whenever $\mathcal{Q}$ is absolutely continuous with respect to $\mathcal{P}$ ($\mathcal{Q} \ll \mathcal{P}$ means for every $\mathcal{P}$ measurable set A, $\mathcal{P}(A) = 0$ means $\mathcal{Q}(A) = 0$), $D(\mathcal{Q} \parallel \mathcal{P}) = \mathrm{Ent}(d\mathcal{Q}/d\mathcal{P})$.

idea of the transportation method

Recall in Section 4.9, we have a duality formula of entropy in **Theorem 4.13**. And it has a corollary:

Corollary 1.

Let Z be a real-valued integrable random variable. Then for every $\lambda \in \mathbb{R}$,

$$\log \mathbb{E}[e^{\lambda(Z-\mathbb{E}[Z])}] = \sup_{\mathcal{Q} \ll \mathcal{P}} [\lambda(\mathbb{E}_{\mathcal{Q}}[Z] - \mathbb{E}[Z]) - D(\mathcal{Q} \parallel \mathcal{P})]$$

where the supremum is taken over all probability measures $\mathcal{Q}$ absolutely continuous with respect to $\mathcal{P}$, and $\mathbb{E}_{\mathcal{Q}}[]$ denotes integration with respect to the measure $\mathcal{Q}$ (recall that $\mathbb{E}$ is integration with respect to $\mathcal{P}$).

The logarithmic moment-generating function of a real-valued random variable Z is denoted by $\psi_Z(\lambda) = \log \mathbb{E}[e^{\lambda} Z]$ for $\lambda \in \mathbb{R}$.

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basic setting

Let $X_1, \ldots, X_n$ be independent random variables taking values in a measurable set $\mathcal{X}$ and consider a measurable function $f : \mathcal{X}^n \to \mathbb{R}$ of n variables. $Z = f(X_1, \ldots, X_n)$. Let $d : \mathcal{X} \times \mathcal{X} \to [0, +\infty)$ be a nonnegative function (a pseudo-metric) and let $c_1, \ldots, c_n \geqslant 0$ be constants. We assume that f satisfies the Lipschitz-type property

$$f(y) - f(x) \leqslant \sum_{i=1}^{n} c_i d(x_i, y_i)$$

for all x, $y \in \mathcal{X}^n$.

Our Goal: prove **Gaussian concentration inequality**:

Theorem 2 (GAUSSIAN CONCENTRATION INEQUALITY).

Let $X = (X_1, \ldots, X_n)$ *be a vector of* n *independent standard normal variables. Let* $f : \mathbb{R}^n \to \mathbb{R}$ *denote an* L*-Lipschitz function. Then, for all* $t > 0$*,*

$$P\{f(X) - \mathbb{E}[f(X)] \geqslant t\} \leqslant e^{-t^2/(2L^2)}.$$

We already know we can achieve this theorem by bounding $\psi_{Z-\mathbb{E}_{\mathcal{P}}[Z]}(\lambda) = \log \mathbb{E}_{\mathcal{P}}[e^{\lambda(Z-\mathbb{E}_{\mathcal{P}}[Z])}]$, where $Z := f(X)$. We can use **the transportation method** to bound it.

the transportation method

Method: transportation lemma.

Lemma 1.

Let Z be real-valued r.v. on $(\Omega, \mathcal{A}, \mathcal{P})$. $\psi_{Z-\mathbb{E}_{\mathcal{P}}[Z]}(\lambda) = \log \mathbb{E}_{\mathcal{P}}[e^{\lambda(Z-\mathbb{E}_{\mathcal{P}}[Z])}] \leqslant \frac{\nu\lambda^2}{2}$ for every $\lambda > 0$ for some $\nu > 0$ iff for any probability measure $\mathcal{Q} \ll \mathcal{P}$ and $D(\mathcal{Q} \parallel \mathcal{P}) < \infty$,

$$\mathbb{E}_{\mathcal{Q}}[f] - \mathbb{E}_{\mathcal{P}}[f] \leqslant \sqrt{2\nu D(\mathcal{Q} \parallel \mathcal{P})}. \tag{*}$$

To prove the inequality (*), we use "coupling". The joint distribution $\boldsymbol{P}$ of the pair (X, Y) is a "coupling" of $\mathcal{P}$ and $\mathcal{Q}$. $\mathbb{P}(\mathcal{P}, \mathcal{Q})$ is the collection of all such distribution. We goes to Kantorovich's formulation of the optimal transportation problem.

Formulation of Optimal Transportation Problem

Consider $\mathcal{P}$ as the joint distribution of $X_1, \ldots, X_n$ on $\mathcal{X}^n$. And Y has distribution $\mathcal{Q}$ on $\mathcal{X}^n$. **Optimal Transportation**:

$$\min_{\boldsymbol{P} \in P(\mathcal{P}, \mathcal{Q})} \mathbb{E}_{\mathcal{P}}[d(X, Y)]$$

which is the "effort" required to transport a mass distributed according to $\mathcal{P}$ into a mass distributed according to $\mathcal{Q}$ measured by the cost function d.

Example 1.

$d(t, t') = \mathbb{1}_{\{t \neq t'\}}$,
$d^2(x, y) = (x - y)^2$ *for* x, $y \in \mathbb{R}^n$.

Now, since

$$\begin{aligned}\mathbb{E}_{\mathcal{Q}}[f] - \mathbb{E}_{\mathcal{P}}[f] &= \mathbb{E}_{\boldsymbol{P}}[f(Y) - f(X)] \\ &\leqslant \sum_{i=1}^{n} c_i \mathbb{E}_{\boldsymbol{P}}[d(X_i, Y_i)] \\ &\leqslant (\sum_{i=1}^{n} c_i^2)^{1/2} (\sum_{i=1}^{n} (\mathbb{E}_{\boldsymbol{P}}[d(X_i, Y_i)])^2)^{1/2},\end{aligned}$$

we need to prove $\min_{\boldsymbol{P} \in P(\mathcal{P},\mathcal{Q})} \sum_{i=1}^{n} (\mathbb{E}_{\boldsymbol{P}}[d(X_i, Y_i)])^2 \leqslant 2 \frac{\nu}{(\sum_{i=1}^{n} c_i^2)} D(\mathcal{Q} \parallel \mathcal{P})$ for some ν.

By a induction in Lemma 8.13, it suffices to prove the inequality for $n = 1$. That is,

$$\min_{\boldsymbol{P} \in P(\mathcal{P},\mathcal{Q})} \mathbb{E}_{\mathcal{P}}[d(X, Y)]$$

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Talagrand's Gaussian Transportation Inequality

Our purpose is to prove the following transportation inequality for the standard Gaussian measure.

Theorem 3.

Let $\mathcal{P}$ be the standard Gaussian probability measure on $\mathbb{R}^n$ and let $\mathcal{Q}$ be any probability measure which is absolutely continuous with respect to $\mathcal{P}$. Then

$$\min_{\boldsymbol{P}\in P(\mathcal{P},\mathcal{Q})} \sum_{i=1}^{n} \mathbb{E}_{\boldsymbol{P}}[X_i - Y_i]^2 \leqslant 2D(\mathcal{Q} \,\|\, \mathcal{P}).$$

This implies the Gaussian concentration inequality (Theorem 2): Assume f is L-Lipschitz function. By Jensen's inequality, for every coupling $\boldsymbol{P}$ of $\mathcal{P}$ and $\mathcal{Q} \ll \mathcal{P}$, one has

$$\mathbb{E}_{\mathcal{Q}}[f] - \mathbb{E}_{\mathcal{P}}[f] = \mathbb{E}_{\boldsymbol{P}}[f(Y) - f(X)] \leqslant L\Big(\sum_{i=1}^{n} \mathbb{E}_{\boldsymbol{P}}[X_i - Y_i]^2\Big)^{1/2} \overset{\text{(Thm. 3)}}{\leqslant} \sqrt{2L^2 D(\mathcal{Q} \,\|\, \mathcal{P})}.$$

From the Lemma 1, $\psi_{Z-\mathbb{E}_{\mathcal{P}}[Z]}(\lambda) \leqslant \frac{L^2\lambda^2}{2}$ for all $\lambda > 0$ where $Z = f(X)$.

Proof of the Talagrand's Gaussian Transportation Inequality

We only need to deal with one-dimensioal case. Then alppy the induction argument of **Lemma 8.13**. Our task remains to find

$$\min_{\boldsymbol{P}\in P(\mathcal{P},\mathcal{Q})} \mathbb{E}_{\boldsymbol{P}}[X-Y]^2$$

Proof.

(omitted) c.f. **Lemma 8.11** and **Lemma 8.12** in the textbook for details. □

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Sub-additivity of entropy

If $Y = f(X_1, \ldots, X_n) \geqslant 0$ then

$$Ent(Y) \leqslant \mathbb{E}[\sum_{i=1}^{n} Ent^{(i)}(Y)]$$